

BINF 6210/8210: Machine Learning for Bioinformatics

Midterm Review

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Midterm Review: Topics

- ▶ Data preprocessing
- ▶ Linear regression
- ▶ Classification
 - ▶ Logistic regression
 - ▶ Classifier evaluation

Focus of the Review

The goal is not simply to memorize formulas, but to understand **what each method does, why we use it, and how to interpret it.**

What Should You Be Able to Do?

By the midterm, you should be able to:

- ▶ Explain concepts in your own words
- ▶ Interpret model parameters and outputs
- ▶ Compare closely related methods
- ▶ Identify inappropriate analysis strategies
- ▶ Explain **why** a method is appropriate or inappropriate
- ▶ Apply concepts to simple biological examples

How Should You Review a Concept?

For each concept, ask yourself:

What is it?



Why do we use it?



How does it work?



How do I interpret it?



What can go wrong?

Data Preprocessing

Why Preprocess Data?

Raw biological data may contain:

- ▶ Different measurement scales
- ▶ Skewed distributions
- ▶ Outliers
- ▶ Missing values
- ▶ Systematic differences among samples
- ▶ Variables with very different variances

Goal

Transform the data into a form appropriate for downstream analysis and modeling.

Four Concepts You Must Distinguish

Operation	Main Purpose
Centering	Shift a variable's location
Scaling	Change the relative magnitude or weighting of variables
Transformation	Change distributional shape
Normalization	Reduce systematic differences among samples

Important

Scaling and normalization address conceptually different problems.

Centering

For a variable x :

$$x' = x - \bar{x}$$

After centering:

$$\bar{x}' = 0$$

What changes?

- ▶ The location of the variable

What does not necessarily change?

- ▶ Variance
- ▶ Distributional shape
- ▶ Relative distances among observations

Autoscaling

Autoscaling:

$$z = \frac{x - \bar{x}}{s}$$

where s is the standard deviation.

After autoscaling:

$$\text{Mean} = 0$$

$$\text{Standard deviation} = 1$$

Conceptual Effect

Variables originally measured on very different scales become more comparable.

Why Can Autoscaling Be Problematic?

Suppose:

- ▶ **Biomarker A**: large biological variation
- ▶ **Biomarker B**: almost constant, with small measurement noise

Autoscaling forces both variables to have:

$$SD = 1$$

If Biomarker B has a very small original standard deviation, dividing by that small value can magnify its small fluctuations.

Key Idea

Equal variance after scaling does not necessarily mean equal biological importance.

Autoscaling vs. Pareto Scaling

Autoscaling

$$x' = \frac{x - \bar{x}}{s}$$

Pareto scaling

$$x' = \frac{x - \bar{x}}{\sqrt{s}}$$

- ▶ Autoscaling removes differences in variance more strongly.
- ▶ Pareto scaling preserves more of the original variance structure.

Scaling vs. Normalization

Problem A

Metabolite concentrations range from 0.01 to 100,000.



Primarily a **variable-scale problem**

Think: Scaling

Problem B

One sample has systematically higher signals because more material was injected.



Primarily a **sample-level systematic difference**

Think: Normalization

Linear Regression

Simple Linear Regression

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Intercept

β_0 is the expected value of Y when $X = 0$.

Slope

β_1 is the expected change in Y associated with a one-unit increase in X .

The model describes the **conditional mean of Y** .

What Does Ordinary Least Squares Do?

For observation i , the residual is:

$$e_i = y_i - \hat{y}_i$$

Ordinary least squares chooses the coefficients that minimize:

$$\sum_i e_i^2$$

Conceptual Interpretation

Find the regression model that minimizes the total squared prediction errors on the training observations.

Multiple Linear Regression

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon$$

Interpretation of β_j :

The expected change in Y associated with a one-unit increase in X_j , **holding all other predictors constant**.

Important

“Holding all other predictors constant” is an essential part of the interpretation.

Why Can Regression Coefficients Change?

Consider a simple model:

$$Y = \beta_0 + \beta_1 X_1 + \epsilon$$

Now add another predictor:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

The interpretation of β_1 has changed.

In the multiple regression model, β_1 asks:

How is X_1 associated with Y among observations having the same value of X_2 ?

Confounding

Suppose age is associated with both exercise and blood pressure:

$$\text{Exercise} \longleftrightarrow \text{Age} \longrightarrow \text{Blood Pressure}$$

A simple regression:

$$\text{Blood Pressure} = \beta_0 + \beta_1(\text{Exercise}) + \epsilon$$

may partly reflect age.

After adding age:

$$\text{Blood Pressure} = \beta_0 + \beta_1(\text{Exercise}) + \beta_2(\text{Age}) + \epsilon$$

the exercise coefficient may change substantially.

Association Is Not Causation

A regression coefficient describes an **association**.

It does not automatically answer:

What would happen if we intervened and changed X ?

Potential issues include:

- ▶ Confounding
- ▶ Omitted variables
- ▶ Selection effects
- ▶ Study design

Logistic Regression

Why Not Use Ordinary Linear Regression?

For a binary outcome:

$$Y \in \{0, 1\}$$

we want:

$$0 < P(Y = 1 \mid X) < 1$$

A linear function can produce values below 0 or above 1.

Logistic regression uses the sigmoid function to map any real-valued linear predictor into the interval:

$$(0, 1)$$

Probability, Odds, and Log-Odds

Probability

$$p = P(Y = 1)$$

Odds

$$\frac{p}{1-p}$$

Log-odds

$$\log\left(\frac{p}{1-p}\right)$$

Probability	Odds	Log-odds
$p = 0.5$	$= 1$	$= 0$
$p > 0.5$	> 1	> 0

Logistic Regression Is Linear in Log-Odds

$$\log \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

Important

Logistic regression is **not linear in probability**.

It is linear in **log-odds**.

Interpreting a Logistic Regression Coefficient

Consider:

$$\log \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 X$$

For a one-unit increase in X :

log-odds change by β_1

Exponentiating gives:

Odds are multiplied by e^{β_1}

Thus:

$$e^{\beta_1}$$

is the **odds ratio** associated with a one-unit increase in X .

Odds Ratio Is NOT Probability Ratio

Suppose:

$$e^{\beta_1} = 2$$

Correct

A one-unit increase in X multiplies the **odds** of $Y = 1$ by 2.

Incorrect

A one-unit increase in X doubles the **probability** of $Y = 1$.

Odds and probability are related by:

$$p = \frac{\text{odds}}{1 + \text{odds}}$$

Therefore, doubling the odds does not generally double the probability.

From Probability to Class Label

Logistic regression produces:

$$\hat{p} = P(Y = 1 \mid X)$$

To produce a class label, choose a threshold t :

$$\hat{Y} = \begin{cases} 1, & \hat{p} \geq t \\ 0, & \hat{p} < t \end{cases}$$

For example, if $t = 0.5$:

$$\hat{p} = 0.72 \Rightarrow \hat{Y} = 1$$

$$\hat{p} = 0.31 \Rightarrow \hat{Y} = 0$$

Important

Probability prediction and class prediction are not the same thing.

Classifier Evaluation

Start with the Confusion Matrix

	Predicted 0	Predicted 1
Actual 0	TN	FP
Actual 1	FN	TP

Before memorizing a formula, ask:

- ▶ Are we starting with **actual positives**?
- ▶ Are we starting with **predicted positives**?
- ▶ Which type of error matters in the application?

Four Core Metrics

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall vs. Precision

Recall / Sensitivity

Of all the **actual positives**, how many did we find?

$$\text{Recall} = \frac{TP}{TP + FN}$$

Precision

Of all the **predicted positives**, how many were actually positive?

$$\text{Precision} = \frac{TP}{TP + FP}$$

Accuracy Can Be Misleading

Suppose we have 1,000 patients:

- ▶ 990 do not have the disease
- ▶ 10 have the disease

A classifier predicts **no disease for everyone**.

Then:

$$TN = 990, \quad TP = 0$$

Accuracy:

$$\frac{990}{1000} = 99\%$$

but:

$$\text{Recall} = 0$$

What Happens When the Threshold Decreases?

Suppose $t : 0.5 \rightarrow 0.3$:

More samples are classified as positive. Some actual positives may change:

$$FN \rightarrow TP$$

Therefore:

Recall generally increases

But some actual negatives may change:

$$TN \rightarrow FP$$

Therefore:

False positives generally increase

and precision may decrease.

The Precision–Recall Tradeoff

Lower threshold



More predicted positives



Usually: Recall ↑ and Precision may ↓

Higher threshold



More selective about calling a sample positive



Usually: Precision ↑ Recall ↓

ROC Curve

As the classification threshold changes, calculate:

$$TPR = \text{Sensitivity} = \text{Recall}$$

and

$$FPR = 1 - \text{Specificity}$$

Then plot:

TPR versus FPR

Key Idea

Each point on the ROC curve corresponds to a particular classification threshold.

Interpreting AUC

AUC = Area Under the ROC Curve.

- ▶ $AUC = 1$: perfect discrimination
- ▶ $AUC = 0.5$: random discrimination
- ▶ $AUC < 0.5$: ranking is effectively reversed

Conceptual Interpretation

AUC is the probability that a randomly selected positive sample receives a higher model score than a randomly selected negative sample.

Precision–Recall Curve

Axes:

$$x = \text{Recall}$$

$$y = \text{Precision}$$

Precision–Recall analysis is especially informative when:

- ▶ The positive class is rare
- ▶ Identifying positive cases is the primary concern

PR Focus

When the model says “positive,” how often is it correct, and how many true positives does it find?

Overfitting and Regularization

Recognizing Overfitting

	Training	Unseen Data
Underfitting	Poor	Poor
Appropriate fit	Good	Good
Overfitting	Very good	Poor

Overfitting

The model learns patterns specific to the training data—including noise—rather than patterns that generalize to new data.

Regularization

Without regularization:

Minimize Loss

With regularization:

Minimize Loss + $\lambda(\text{Penalty})$

Conceptual Tradeoff

Accept a slightly worse fit to the training data in exchange for a simpler model with smaller coefficients that may generalize better.

L1 vs. L2 Regularization

	L1 / Lasso	L2 / Ridge
Penalty	$\sum_j \beta_j $	$\sum_j \beta_j^2$
Shrinks coefficients	Yes	Yes
Can make coefficients exactly zero	Yes	Generally no
Implicit feature selection	Yes	No

What Does C Mean in scikit-learn?

For LogisticRegression:

$$C \propto \frac{1}{\text{regularization strength}}$$

Therefore:

small $C \Rightarrow$ strong regularization

and

large $C \Rightarrow$ weak regularization

Question:

Which model is more strongly regularized?

$$C = 0.01 \quad \text{or} \quad C = 100$$

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$$C = 0.01$$

Generalization and Cross-Validation

Three Different Jobs for Data

Training Data

Learn model parameters

Validation Data

Select models or hyperparameters

Test Data

Estimate final generalization performance

The Test Set Has a Special Role

The test set should:

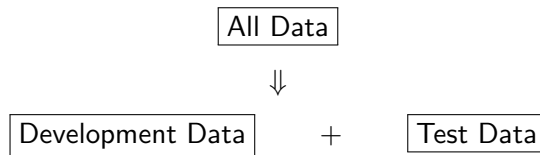
- ▶ Remain untouched during model development
- ▶ Not determine C
- ▶ Not determine which model is selected
- ▶ Not determine feature-selection decisions
- ▶ Be evaluated only after model development is complete

Key Principle

Once the test set influences model selection, it is no longer an independent final evaluation set.

Cross-Validation

First split:



Perform K -fold cross-validation on the **development data only**.

- ▶ Train on $K - 1$ folds
- ▶ Validate on the remaining fold
- ▶ Repeat until every fold has served as validation data
- ▶ Average validation performance

What Is an Out-of-Fold Prediction?

During K -fold cross-validation:

Out-of-Fold Prediction

An out-of-fold (OOF) prediction is a prediction for a sample made by a model that was trained **without that sample**.

For example, in 5-fold CV:

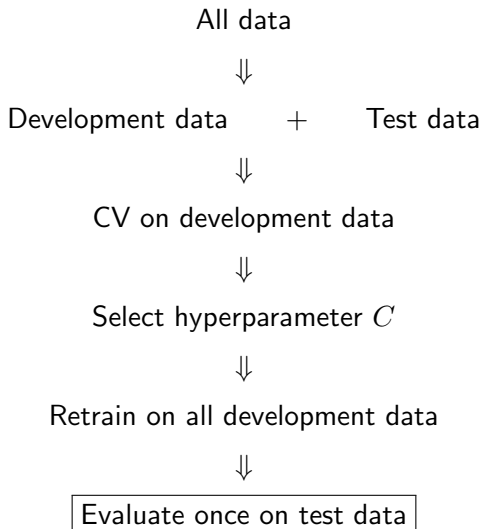
Train on B+C+D+E



Predict samples in A

Those predictions for samples in A are out-of-fold predictions.

Correct Hyperparameter-Selection Workflow



Concept Checks

Concept Check 1

A gene-expression variable has very small variance. After autoscaling, its small random fluctuations become prominent.

Why?

- A. Centering increases its variance
- B. Division by a small SD amplifies the fluctuations
- C. Normalization changes its biological meaning
- D. Logistic regression requires equal variance

Concept Check 1

A gene-expression variable has very small variance. After autoscaling, its small random fluctuations become prominent.

Why?

- A. Centering increases its variance
- B. Division by a small SD amplifies the fluctuations
- C. Normalization changes its biological meaning
- D. Logistic regression requires equal variance

Answer

B. Division by a small SD amplifies the fluctuations.

Concept Check 2

A multiple linear regression gives:

$$\hat{Y} = 5 + 4X_1 - 3X_2$$

What does the coefficient 4 mean?

Concept Check 2

A multiple linear regression gives:

$$\hat{Y} = 5 + 4X_1 - 3X_2$$

What does the coefficient 4 mean?

Answer

Holding X_2 constant, a one-unit increase in X_1 is associated with a 4-unit increase in predicted Y .

Which ideas are essential?

- ▶ **one-unit increase**
- ▶ **associated with**
- ▶ **holding X_2 constant**

Concept Check 3

A logistic regression gives:

$$e^{\beta_1} = 0.5$$

What does this mean?

Concept Check 3

A logistic regression gives:

$$e^{\beta_1} = 0.5$$

What does this mean?

Correct

For each one-unit increase in X_1 , the odds of class 1 are multiplied by 0.5.

Equivalent statement:

The odds are reduced by half.

Not Correct

The probability is reduced by half.

Concept Check 4

A classifier has:

Training accuracy = 98%

Validation accuracy = 70%

What should immediately concern you?

Concept Check 4

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Training accuracy = 98%

Validation accuracy = 70%

What should immediately concern you?

Overfitting

Possible approaches include:

- ▶ Increase regularization
- ▶ Reduce model complexity
- ▶ Reevaluate features or model design

Concept Check 5

A researcher tests 20 values of C on the **test dataset** and reports the best test accuracy.

What is wrong?

Concept Check 5

A researcher tests 20 values of C on the **test dataset** and reports the best test accuracy.

What is wrong?

The test data have participated in:

Model selection

Therefore:

The final test estimate is no longer independent

Hyperparameters should instead be selected using validation data or cross-validation within the development data.

Five Distinctions to Know Cold

1. **Scaling vs. Normalization**
2. **Association vs. Causation**
3. **Probability vs. Odds**
4. **Precision vs. Recall**
5. **Validation Data vs. Test Data**

A Useful Self-Test

If you can explain each pair clearly to another student **without looking at your notes**, you are well prepared for the conceptual portion of the midterm.

Final Review Checklist

Before the midterm, make sure you can explain:

- ▶ Why and when preprocessing is needed
- ▶ How to interpret linear regression coefficients
- ▶ Why association does not imply causation
- ▶ Probability, odds, and log-odds and how to interpret an odds ratio
- ▶ Accuracy, precision, recall, and specificity
- ▶ How classification thresholds affect performance
- ▶ ROC vs. Precision–Recall curves
- ▶ Overfitting and regularization, L1 vs. L2 regularization
- ▶ The meaning of C
- ▶ Training vs. validation vs. test data
- ▶ How cross-validation is used for model selection

Questions?

Good luck on the midterm!